

Azure Load Balancer Outbound Rules and SNAT Planning – Architecture

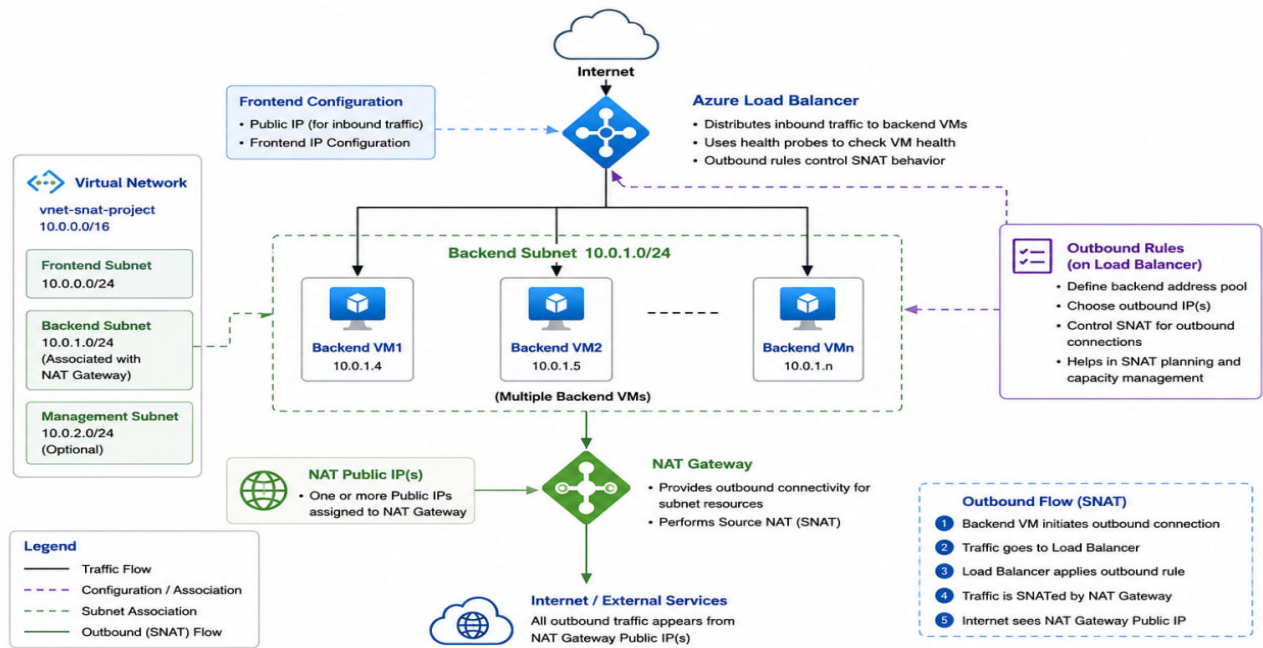


Figure 1: Azure Load Balancer and NAT Gateway architecture

Azure Load Balancer Outbound Rules and SNAT Planning

The architecture is best understood as two separate directions of traffic. Inbound traffic flows from the Internet through the Load Balancer to the backend VMs. Outbound traffic flows from the backend VMs through the NAT Gateway (using SNAT) back out to the Internet. These two paths are handled by different components and never overlap once traffic leaves the backend subnet.

Inbound path - Load Balancer

The Load Balancer has a frontend IP, which is the single public address clients connect to. It distributes incoming requests across a backend pool of VMs (for example VM1 at 10.0.1.4 and VM2 at 10.0.1.5). A health probe continuously checks each VM, and only healthy VMs are sent new traffic. This setup improves availability and lets the application scale across multiple instances.

Networking layer

All resources sit inside one virtual network, vnet-snat-project, with address space 10.0.0.0/16. The VMs are placed inside a dedicated backend-subnet (10.0.1.0/24). The NAT Gateway is associated with this subnet, which is what gives every VM in it outbound connectivity.

Outbound path - SNAT via NAT Gateway

A VM's private IP, such as 10.0.1.4, cannot be used directly on the public Internet. The NAT Gateway (nat-snat-project) solves this using SNAT (Source Network Address Translation): it rewrites the source address of outbound connections to its public IP, 48.223.130.231, before the traffic leaves Azure. So before translation the source is 10.0.1.4, and after translation the source becomes 48.223.130.231 - the external service only ever sees the public IP.

- SNAT ports keep many simultaneous outbound connections distinguishable from each other.
- High connection volume can use up all available SNAT ports, causing SNAT port exhaustion - seen as failed outbound connections, timeouts, or intermittent errors.
- SNAT planning means estimating outbound capacity in advance, based on the number of VMs, expected connection volume, number of public IPs on the NAT Gateway, and how efficiently the application reuses connections.

Verified test

From vm-snat-01 (private IP 10.0.1.4), running `curl -4 https://api.ipify.org` returned 48.223.130.231, confirming that outbound traffic is correctly translated through the NAT Gateway's public IP.